

REQUIREMENT ANALYSIS

2.1 Functional Requirements (Approvals Focus)

Functional requirements define the explicit execution behaviors and database actions that the ServiceNow Flow Designer engine must perform to ensure process compliance across the corporate domain.

- **FR-APP-01: Auto-Trigger on Catalog Submission**

The workflow engine must initiate execution immediately when an item matching the unique identifier of the Laptop Request Catalog Form is created in the database.

- **FR-APP-02: User Profile Relation Parsing**

The flow must query the User record table (sys_user) to pull the active manager field string associated with the specific requested_for profile data pill.

- **FR-APP-03: System Approval Record Creation**

The engine must inject a structural approval task into the sysapproval_approver table, setting its state to *Requested* and binding it to the parent Requested Item (RITM) record.

- **FR-APP-04: Logical Fork Monitoring**

The workflow engine must halt sequence execution at the approval checkpoint, waiting continuously for the status variable to update before evaluating the conditional engine parameters.

2.2 Non-Functional Requirements

Non-functional requirements govern the platform operating boundaries, security constraints, and systemic validation checks required to maintain instance integrity.

- **NFR-APP-01: Performance Execution Ceiling**

The lookup queries and record creation steps inside the dynamic execution engine must resolve within 400 milliseconds of catalog entry initialization.

- **NFR-APP-02: Relational Data Security**

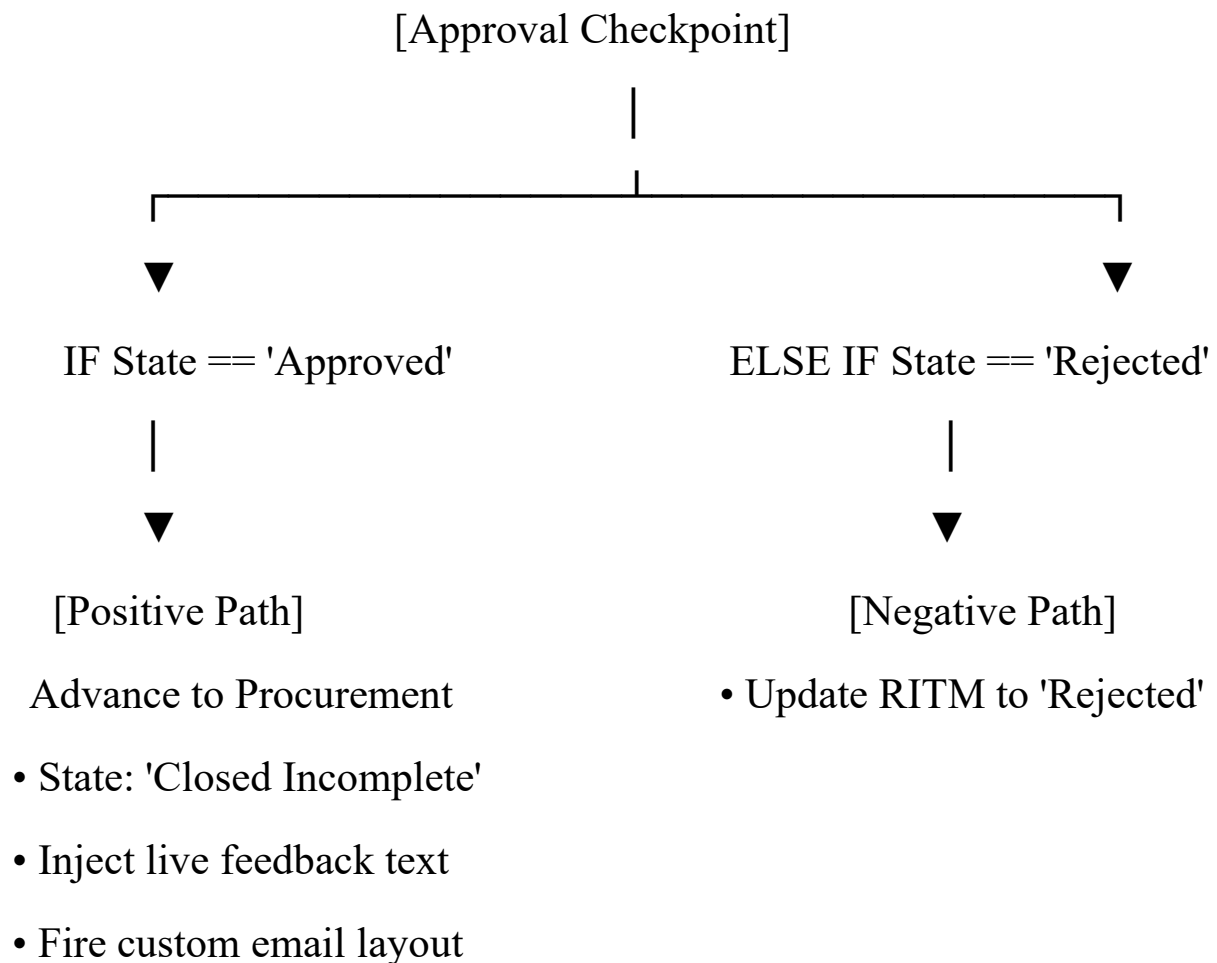
Access to write or alter approval decision records must be locked via Access Control Lists (ACLs), ensuring only the explicit manager or system administrators can process updates.

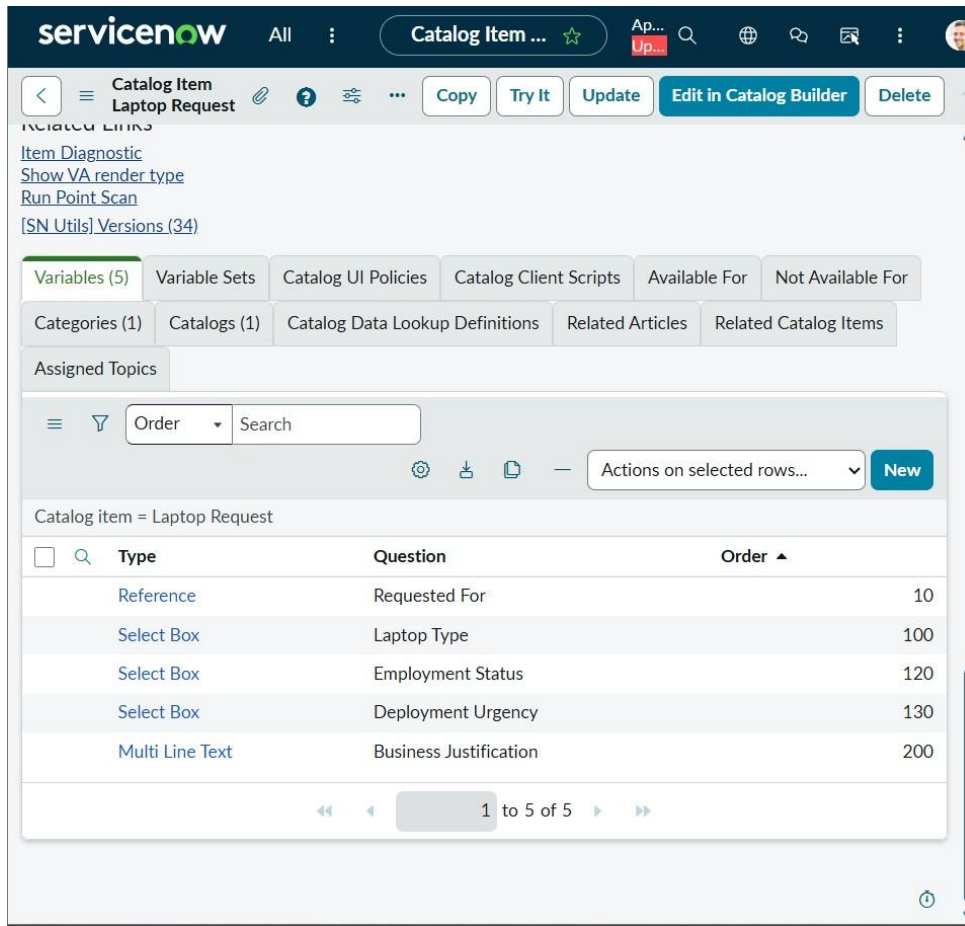
- **NFR-APP-03: HTML Layout Compatibility**

Communication structures sent by the platform must use standardscompliant HTML elements, ensuring layout consistency across various mobile devices and web mail interfaces.

2.3 Logic Matrix & Workflow Decision Tree

The business rules governing the Phase 1 deployment require a strict conditional framework. Under no circumstances should any subsequent fulfillment logic or system assignment occur prior to the manager variable resolving to an explicit approved state.





2.4 User Persona Mapping for Approvals

The processing architecture of the automated system interfaces with two distinct corporate roles during the execution of Phase 1:

-  **The Submitting Employee (Requester):** Needs to see transparent progress indicators on the Service Portal interface showing that their ticket is currently pending authorization from their respective supervisor.
-  **The Corporate Supervisor (Approver):** Needs a highcontext view summarizing request specifics—including laptop model, item cost, and user justifications—accompanied by actionable link items to submit a decision quickly.

2.5 Engineering Requirement Matrix

Requirement Code	Engineering Category	Technical Description Boundary	Target Evaluation Metric
REQ-APP-F1	Functional Logic	Dynamic parent record supervisor extraction query.	Zero empty routing errors across active profiles.
REQ-APP-S1	Data Governance	Restricting change access on active authorization states.	Unauthenticated users are blocked from editing items.
REQ-APP-N1	User Interface	Embedding live manager feedback text strings into layouts.	Rejection records render system data accurately.